

Abstract No.

From Diversity to Dependency: Plasticity of Myeloid Cells in Brain Tumors

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Glioblastoma evolves within a chronically damaged, metabolically constrained brain microenvironment, co-opting resident microglia and infiltrating macrophages to sustain tumor growth and evade therapy. How spatial tissue context and metabolic stress drive microglial reprogramming and myeloid heterogeneity in glioblastoma is a central question. Using spatial transcriptomics, multi-omics profiling, and functional perturbation approaches in preclinical models and patient datasets, the spatiotemporal organization of the glioblastoma myeloid compartment reveals regionally defined microglial states that emerge along tumor invasion fronts and necrotic niches. Among these, a lipid-laden, GPNMB-expressing macrophage population, enriched in metabolically stressed tumor regions, actively recycles myelin lipids and transfers them to glioblastoma cells, fueling mesenchymal malignancy. These spatially constrained myeloid states suppress anti-tumor immunity and drive therapeutic resistance. Defining how brain-specific spatial cues govern microglial identity and function in glioblastoma uncovers niche-specific vulnerabilities that can be leveraged to reprogram the tumor microenvironment and restore durable therapeutic responses



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